

KHALKGOL, Mongolia

WMO: 443130

Lat: 47.6300N

Lon: 118.6222E

Elev: 688

StdP: 93.33

Time Zone: 8.00 (E08)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.4	-34.7	-39.2	0.1	-37.3	-36.5	0.1	-34.5	11.6	-17.7	9.5	-19.0	1.0	180	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	32.5	19.1	30.3	18.3	28.4	17.6	21.9	28.3	20.5	26.4	19.3	25.0	5.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.0	15.9	24.7	18.6	14.6	23.2	17.4	13.5	22.1	68.0	28.8	62.5	26.4	58.4	24.9	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	9.4	8.7	DB	-40.8	35.8	2.7	2.5	-42.8	37.6	-44.3	39.1	-45.8	40.4	-47.8	42.3	(4)
(5)			WB	-40.9	23.2	2.8	2.0	-42.9	24.6	-44.6	25.8	-46.2	26.9	-48.2	28.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.3	-24.0	-19.5	-9.4	3.2	11.5	18.2	21.1	19.0	12.0	2.1	-11.0	-20.8	(6)	
	DBStd	16.72	5.72	6.32	7.33	5.24	4.53	3.90	2.96	3.41	4.51	5.20	6.73	5.69	(7)	
	HDD10.0	4589	1053	826	601	212	35	0	0	1	30	249	630	954	(8)	
	HDD18.3	6777	1311	1059	858	454	217	49	7	31	196	502	880	1212	(9)	
	CDD10.0	1053	0	0	0	8	82	246	344	279	89	5	0	0	(10)	
	CDD18.3	199	0	0	0	0	6	44	93	51	5	0	0	0	(11)	
	CDH23.3	2358	0	0	0	7	137	613	934	562	103	2	0	0	(12)	
	CDH26.7	815	0	0	0	1	41	227	338	184	25	0	0	0	(13)	
(14) Wind	WSAvg	3.3	2.6	2.7	3.3	4.2	4.2	3.5	3.2	3.2	3.5	3.5	2.9	2.6	(14)	
(15) Precipitation	PrecAvg	299	2	2	3	11	16	43	96	66	31	12	5	3	(15)	
	PrecMax	598	12	8	12	59	46	150	211	194	91	46	18	11	(16)	
	PrecMin	105	0	0	0	0	2	1	21	10	0	0	0	0	(17)	
	PrecStd	110	2	2	3	11	10	34	47	41	24	11	5	3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.2	1.2	13.1	24.0	30.9	34.8	35.5	34.4	29.8	21.7	8.9	-3.0	(19)	
		MCWB	-8.5	-2.3	5.5	11.0	15.1	18.4	22.2	21.0	16.4	10.9	3.2	-4.7	(20)	
	2%	DB	-10.0	-2.6	7.8	19.2	27.2	31.6	32.7	30.8	26.4	17.1	4.7	-7.5	(21)	
		MCWB	-11.0	-4.5	2.4	8.4	14.0	17.6	20.2	19.7	14.8	8.1	0.7	-8.5	(22)	
	5%	DB	-12.5	-6.0	4.5	16.0	24.0	29.3	30.3	28.6	23.6	13.9	1.8	-9.9	(23)	
		MCWB	-13.1	-7.3	0.2	7.2	12.4	17.0	18.9	18.3	13.6	7.0	-0.9	-10.7	(24)	
	10%	DB	-15.0	-9.0	1.6	12.9	21.1	27.0	28.2	26.5	20.9	11.0	-1.0	-12.0	(25)	
		MCWB	-15.5	-10.0	-1.4	5.5	11.2	16.0	18.6	17.4	12.5	5.6	-2.9	-12.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.2	-1.7	5.8	12.8	17.1	21.5	24.5	24.3	18.8	12.2	4.0	-4.6	(27)	
		MCDB	-7.1	1.0	12.4	20.4	25.9	28.7	31.8	31.0	25.1	18.8	8.9	-2.7	(28)	
	2%	WB	-10.9	-4.5	2.7	9.5	15.0	19.5	22.3	21.8	16.6	9.2	1.5	-8.4	(29)	
		MCDB	-10.1	-2.4	7.2	18.4	24.3	27.1	28.8	28.2	22.8	16.4	5.1	-7.6	(30)	
	5%	WB	-13.1	-7.4	0.6	7.4	13.5	18.3	21.0	20.0	15.2	7.5	-0.3	-10.6	(31)	
		MCDB	-12.5	-6.1	4.0	15.0	22.2	25.7	27.0	25.4	21.0	12.8	2.5	-10.0	(32)	
	10%	WB	-15.5	-10.0	-1.3	5.7	12.0	17.3	19.9	18.7	13.7	6.0	-2.0	-12.6	(33)	
		MCDB	-15.0	-9.1	1.5	12.2	20.0	24.3	25.4	24.1	19.0	10.7	0.3	-12.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	13.5	13.3	12.7	13.8	13.4	11.6	12.3	13.4	12.4	11.4	11.2	(35)	
		MCDBR	13.1	15.6	15.3	18.2	19.4	17.3	15.8	16.1	17.8	17.2	14.3	11.7	(36)	
	5% WB	MCWBR	12.3	14.0	10.5	10.3	9.3	6.9	6.1	6.7	8.4	9.9	9.9	10.9	(37)	
		MCWBR	13.0	15.3	14.7	16.9	16.8	13.9	12.5	12.9	14.4	15.8	13.7	11.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.263	0.285	0.335	0.407	0.485	0.480	0.437	0.402	0.356	0.325	0.292	0.270	(40)	
		taud	2.229	2.178	2.122	2.043	1.875	2.013	2.211	2.279	2.340	2.327	2.262	2.199	(41)	
	Ebn at Noon		815	872	874	840	784	792	824	839	849	816	765	746	(42)	
		Edh at Noon	78	103	129	156	192	168	136	122	104	89	76	71	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.96	3.22	4.67	5.46	6.13	6.23	5.88	5.43	4.38	3.09	1.97	1.51	(44)	
	RadStd		0.12	0.18	0.24	0.35	0.39	0.46	0.41	0.38	0.33	0.21	0.11	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	-2.73	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page